# Junior Deogracias MAMBO MOTSOU

## Embedded Software Engineer

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## Profile

Embedded software engineer (Polytech Dijon, 2026) focused on C firmware, real-time operating systems and their tooling. At FORVIA I wrote and debugged firmware for automotive microcontrollers and a multithreaded Python/Kotlin control stack for a lighting demonstrator. In personal projects I build an RTOS, an x86-64 OS and a C toolchain, with attention to build systems, CI and testing.

## Technical skills

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|---|---|
| **Languages** | C (C99/C11), Python, C++ (Qt), Kotlin; assembly: x86-64 (NASM), Arm Thumb, RISC-V |
| **RTOS & OS** | FreeRTOS (tasks, queues, mutexes, event groups, stream buffers), Zephyr (devicetree, Kconfig), bare-metal; kernel internals: scheduling, context switch, paging, interrupts, syscalls, filesystems |
| **Linux** | Ubuntu as daily host; user-space drivers (spidev, i2c-dev, sysfs GPIO) on Raspberry Pi |
| **Build & toolchains** | GCC (arm-none-eabi, x86_64-elf cross), Clang, linker scripts, CMake, Ninja, GNU Make, West, Kconfig; IAR EWARM, MPLAB X (XC16/XC32) |
| **Debug & test** | GDB (remote, QEMU), J-Link, QEMU; unit tests with ASan/UBSan, conformance and differential testing against reference models |
| **Automation** | Git (pull-request workflow), GitHub Actions CI, Python tooling |
| **Interfaces** | UART, SPI, I²C, LIN, MeLiBu, Bluetooth serial with a JSON protocol, REST/HTTP clients |
| **Applications** | Python multithreading (PyQt5, OpenCV), Qt/C++, Android (Kotlin, Jetpack Compose) |

## Professional experience

### Embedded Systems Engineer — apprentice (contrat de professionnalisation)

Sep. 2025 – Sep. 2026

FORVIA – Faurecia Interior Systems Surface Activation department, France

**Interior ambient-lighting demonstrator (software owner)**

- Extended the control software of an automotive interior demonstration buck from a few LED displays to nine STM32 UART gateways driving WS2812 LED strips across several lighting zones.
- Wrote a multithreaded Python/PyQt5 application: OpenCV decodes MP4 animations, maps logical LED IDs to frame pixels and distributes each frame to one worker thread per gateway at 921 600 baud, with a barrier to keep all nine gateways in step; custom serial frames (ASCII with delta compression).
- Built an Android tablet app (Kotlin, Jetpack Compose) that controls zones, colours and brightness presets over Bluetooth serial using a JSON command protocol with explicit error codes; added a keyboard fallback mode.
- Adapted the STM32 gateway firmware with minimal changes (ID-offset filtering when two gateways share a UART line).

Environment: Python 3, PyQt5, OpenCV, Kotlin/Jetpack Compose, STM32 (C), WS2812, UART, Bluetooth SPP, Git

**Firmware bring-up: LIN-controlled heating-element controller (TCU)**

- Developed C firmware on the Infineon TLD4020-3ET (Arm Cortex-M23 with integrated LIN): LIN ON/OFF command handling, heater PWM through the LDRV peripheral, NTC-based temperature regulation, fault protection and a state-machine architecture.
- Diagnosed early bring-up blockers: traced a HardFault in device initialisation to a silicon-revision mismatch on the evaluation board (register absent on older silicon), and a lost debug connection to boot-mode jumper and USB power conditions.
- The NTC divider is supplied from the switched heater output, so the ADC node is a 100 Hz square wave during PWM. Synchronised sampling to the PWM on-time (ISR at mid-pulse, 16-sample averaging), which brought measurement scatter within the regulation requirement.

Environment: TLD4020-3STD_EVAL board, IAR Embedded Workbench, Infineon MCU Configuration Wizard, on-board J-Link, printf over LIN (MobaXterm, 19 200 baud 8N1)

**MeLiBu LED driver port: Melexis MLX81116 to MLX81143**

- Ported an existing MeLiBu 1.x driver on RP2040 to the MLX81143: extended-mode sub-address encoding (14 LEDs per node), linear 15-bit PWM encoding confirmed against the datasheet and on the hardware, for a 22 × 7 panel (154 RGB LEDs).
- Fixed bus-timing faults: break generation with busy-wait while interrupts are masked, critical section narrowed to the break to stop UART RX byte loss, TX-FIFO drain to prevent back-to-back frame collisions; corrected the bus return pin.

Environment: RP2040, Raspberry Pi Pico SDK, PIO state machines, C, MeLiBu 1.x

### Embedded Systems Intern

Apr. – Aug. 2025

KMUTT — MDT Lab Bangkok, Thailand

- Image-processing work for a precision-agriculture project: fruit classification by transfer learning on VGG16 (Keras/TensorFlow), 98.4 % accuracy on a 4 599-image test set (Fruits-360, 131 classes), plus an OpenCV contour/colour detection pipeline.

Environment: Python, TensorFlow/Keras, OpenCV

## Personal & academic projects

### Personal

Ongoing. I define the architecture, specifications and verification strategy; implementation is assisted by AI coding tools. Validation so far is in emulation (QEMU, EmbSIM), not on physical boards.

**EmbLinkRTOS — preemptive RTOS for microcontrollers** personal — Oct. 2026 –
github.com/EmbLinkOs/EmbLinkRTOS

- Fixed-priority preemptive scheduler (bitmap ready lists, FIFO within a priority); Cortex-M port with PendSV/PSP context switch, lazy FPU save and BASEPRI critical sections; RISC-V (mret, CLINT) and AVR ports.
- Semaphores, mutexes with priority inheritance and priority ceiling, timeouts; conformance tests and differential testing against a Python reference model of the scheduler.
- CMake/Kconfig build with board definitions (NUCLEO-F446RE, Arduino Uno, ESP32-C3); GitHub Actions CI with ASan/UBSan/TSan, QEMU and EmbSIM runs.

**EmbLinkOS — x86-64 operating system from scratch** personal — May 2026 –
github.com/EmbLinkOs/EmbLinkOs

- Two-stage BIOS bootloader in NASM (real, protected, then long mode) and a UEFI PE32+ loader, higher-half kernel in C: 4-level paging, APIC/IO-APIC, SMP, ring-3 processes, syscalls, ELF loader.
- Copy-on-write B-tree filesystem (CRC32C, snapshots, AES-XTS) checked against a Python oracle; FAT32, VFS; PCI, AHCI, NVMe and USB drivers; newlib and TCC ports.
- In-kernel selftests, host unit tests under ASan/UBSan, QEMU-driven integration tests; GDB remote debugging.

**EmbCC — C cross-compiler toolchain** personal — Jul. 2026 –
github.com/EmbLinkOs/EmbCC

- C99 compiler, assembler and linker with Cortex-M, RISC-V, AVR and x86-64 back-ends; firmware utilities (SVD-to-header/startup generation, HEX/UF2 packing).
- Verification: encodings cross-checked against llvm-mc/GNU as, generated code run under QEMU and compared with GCC/Clang; GitHub Actions CI on Linux and macOS with GCC and Clang.

**RFID/PIN access-control terminal on FreeRTOS** personal — Oct. 2026
github.com/teo1747/stm32-freertos-access-control

- Register-level drivers (no vendor HAL) on STM32F446RE: MFRC522 RFID on SPI, SSD1306, DS3231 RTC and 24LC256 EEPROM on I²C, keypad, servo PWM, watchdog; six FreeRTOS tasks with queues, mutexes, event groups and stream buffers; salted, stretched SHA-256 PINs; CRC-checked log.
- Host unit tests with sanitizers and system scenarios in EmbSIM; not yet tested on a physical board.

Environment: NUCLEO-F446RE, FreeRTOS 11, CMSIS, arm-none-eabi-gcc, CMake, Ninja

### Polytech Dijon

**Real-time thermometer: bare-metal PIC24, then FreeRTOS on PIC32MZ** Polytech, solo then pair — 2025 – 2026
github.com/teo1747/pic24-32-rt-thermometer

- V1 (solo): bare-metal PIC24FJ1024GB610 with RTCC timestamps and SPI EEPROM logging. V2 (pair): FreeRTOS pipeline of acquisition, analysis and display tasks on PIC32MZ2048EFH100, linked by queues and binary semaphores; timestamped messages; °C/°F toggle on a button.

Environment: Explorer 16/32, MPLAB X, XC16/XC32, Harmony v3/MCC, FreeRTOS 10.5, TC1047A sensor, on-board debugger

**Zephyr RTOS feasibility study on STM32** Polytech lab — 2025 – 2026
github.com/teo1747/zephyr-stm32-sensor-hub

- Three threads share one I²C bus: BMP280 (temperature/pressure), MPU6050 (6-axis IMU) and an SSD1306 OLED through the display API and CFB; shared data protected by a mutex; board configured through a devicetree overlay. (Repository rebuilt from the lab report.)

Environment: NUCLEO-G474RE, Zephyr, West, CMake, Ninja, STM32CubeProgrammer, serial console

**Embedded Linux peripherals on Raspberry Pi** Polytech, team — Mar. 2025

- User-space C drivers for Pmod modules: SPI ambient-light sensor through spidev (mode 3, 2 MHz), I²C humidity sensor through i2c-dev, GPIO through sysfs; built with Make.

## Education & languages

**Engineering degree (Diplôme d'ingénieur, Master's level) — Embedded Systems** — 2023 – Sep. 2026
Polytech Dijon Dijon, France
Graduated. Final-year apprenticeship at FORVIA. Coursework: electronics, PCB design, real-time systems, SoC/FPGA.

**Two-year preparatory programme — mathematics, physics, electronics** — 2021 – 2023
IUC Cameroon

**Languages** English: native (anglophone), TOEIC C1 (2024) | French: fluent, professional use (B2)
**Certification** Cisco CCNA 1 (July 2024)